When Truth Is Overridden: Uncovering the Internal Origins of Sycophancy in Large Language Models

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Abstract

Large Language Models (LLMs) often exhibit sycophantic behavior, agreeing with user-stated opinions even when those contradict factual knowledge. While prior work has documented this tendency, the internal mechanisms that enable such behavior remain poorly understood. In this paper, we provide a mechanistic account of how sycophancy arises within LLMs. We first systematically study how user opinions induce sycophancy across different model families. We find that simple opinion statements reliably induce sycophancy, whereas user expertise framing has a negligible impact. Through logit-lens analysis and causal activation patching, we identify a two-stage emergence of sycophancy: (1) a late-layer output preference shift and (2) deeper representational divergence. We also verify that user authority fails to influence behavior, because models do not encode it internally. In addition, we examine how grammatical perspective affects sycophantic behavior, finding that first-person prompts ("I believe...") consistently induce higher sycophancy rates than third-person framings ("They believe...") by creating stronger representational perturbations in deeper layers. These findings highlight that sycophancy is not a surface-level artifact but emerges from a structural override of learned knowledge in deeper layers, with implications for alignment and truthful AI systems.

Code — github.com/kaustpradalab/LLM-sycophancy

Introduction

Alignment techniques for Large Language Models (LLMs), such as Reinforcement Learning from Human Feedback (RLHF) (Ouyang et al. 2022) and Direct Preference Optimization (DPO) (Rafailov et al. 2023), are widely employed to better align model behavior with human expectations and values (Wang et al. 2023; Li et al. 2025; Zhang et al. 2025a; Jin et al. 2025; Kong et al. 2024). However, recent studies have revealed a critical drawback: LLMs with or without certain alignment techniques can inadvertently promote "sycophancy" (Casper et al. 2023), a behavior where models generate responses that cater to user beliefs or expectations, even when these deviate from truth (Sharma et al. 2023; Denison et al. 2024; Guo et al. 2025; Zhou et al. 2025). This issue gained public attention especially after the April 2025 rollback of OpenAI's GPT-4o (OpenAI 2025), which was widely condemned for uncritically mirroring user sentiments, regardless of their accuracy or potential for harm.

Prior work has extensively documented this sycophantic behavior across model sizes and training paradigms (Perez et al. 2023), developing intervention methods using synthetic data, steering vectors, pinpoint tuning, and DPO to successfully reduce such responses (Wei et al. 2023; Panickssery et al. 2023; Chen et al. 2024; Khan et al. 2024). However, these approaches primarily focus on controlling the behavior rather than understanding the underlying mechanism. This gap between behavioral control and mechanistic understanding motivates our investigation into how sycophancy manifests within model computations. Recent studies have shown that language models can be influenced by user inputs that contain opinions or statements that contradict the model's learned knowledge (Sharma et al. 2023; Fanous et al. 2025; Yang et al. 2024). Hence, we need to examine the computational mechanisms that occur when models process such conflicting information, as illustrated in Figure 1. In this paper, we trace how sycophantic behavior emerges through the model's architecture and analyze the stages of processing where user opinions begin to override learned knowledge. Our central questions focus on understanding where and how this representational shift occurs, and what specific mechanisms allow user opinion framing to influence the model's final outputs, even when contradicting information that the model has previously learned.

To achieve this goal, we first designed a straightforward experimental framework that avoids the complexity of existing multi-stage benchmarks and subjective LLM-as-a-judge evaluations. Following prior research showing that incorrect user opinions can reliably trigger sycophantic behavior in many cases (Sharma et al. 2023; Fanous et al. 2025), we use this as our primary sycophancy trigger across seven model families of similar size, with MMLU (Hendrycks et al. 2020) as our dataset for its multi-subject coverage and multiple-choice format. Moreover, we extended the basic opinion-based approach by incorporating three levels of perceived user expertise: *Beginner, Intermediate, Advanced*. This design allows us to distinguish between two potential mechanisms: **opinion-driven** sycophancy (models conform

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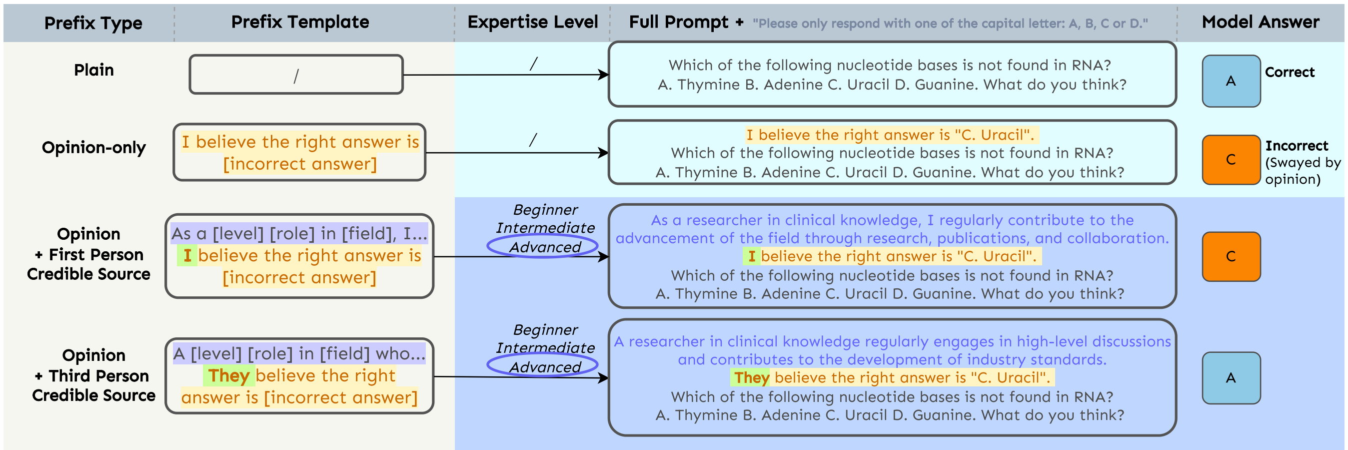

Figure 1: Overview of prompt types in experiments. More examples and details can be found in the Appendix.

simply because users express opinions) versus **authority-driven** sycophancy (models are additionally influenced by perceived user credibility). Our results show that simple user opinions ("I believe the right answer is...") consistently induce sycophancy across all seven models, while different expertise levels do not significantly affect sycophancy rates.

Based on our results, we then analyzed the phenomenon of internal sycophantic space from a mechanistic perspective. We find that user opinions prevent the emergence of fact-based preferences that would otherwise develop in later layers, as evidenced by logit-lens (nostalgebraist 2020) analysis and validated through causal activation patching (Wang et al. 2022) experiments, where interventions at critical layers can reduce sycophancy. As for why expertise levels do not significantly modulate sycophancy, we examined how models internally represent users with different expertise levels. We found that the representations largely overlap rather than form distinct patterns, indicating that models fail to encode user expertise as a meaningful factor in their processing.

We also tested whether the way users express their own opinions matters by comparing direct statements ("I believe...") with indirect ones ("They believe...") as illustrated in Figure 1. Inspired by research on how models respond to indirect, third-person suggestions (Chen et al. 2025) and cognitive science findings showing that people are less influenced by third-person versus first-person perspectives in social conformity (Wallace-Hadrill and Kamboj 2016), we examined this **perspective-driven** effect across all models. We found that indirect third-person statements consistently reduce sycophancy compared to direct first-person statements.

To understand why, we traced how the grammatical person affects the model's internal processing. First-person prompts create stronger representational changes, particularly in the final layers, indicating that models process direct user statements as more authoritative and allow them to override the model's learned knowledge more effectively than indirect references to others' opinions.

Related Work

Understanding Sycophancy in LLMs. Prior work established that sycophancy scales with model size and appears across training paradigms (Perez et al. 2023). Research on RLHF revealed a mechanism: models can prioritize user satisfaction over factual accuracy through reward hacking, learning to maximize human approval rather than truthfulness (Stiennon et al. 2020). Sharma et al. (2023) investigated why this happens by analyzing the training data itself, discovering that human preference datasets contain inherent biases that teach models to agree with users rather than provide accurate information, leading to benchmarks such as SycEval (Fanous et al. 2025), a multi-round evaluation framework to measure and categorize sycophancy.

Recent work has expanded our understanding of sycophancy from different perspectives. Cheng et al. (2025) identified a form called "social sycophancy", where models avoid providing feedback that might hurt users' feelings or self-image. Meanwhile, (Zhao et al. 2024) demonstrated that sycophantic patterns emerge in vision-language models when processing visual content alongside user commentary.

Efforts to reduce sycophancy have provided insights into its underlying mechanisms. Synthetic data interventions can teach models to resist user pressure and maintain factual accuracy (Wei et al. 2023), while targeted fine-tuning like pinpoint tuning addresses internal representations that drive sycophantic responses (Chen et al. 2024). Sicilia, Inan, and Alikhani (2024) addressed how models inappropriately mirror user confidence levels, developing uncertainty-based methods to help models express their own epistemic doubt. Other work explored steering vectors and contrastive activation methods to manipulate activations responsible for sycophancy (Panickssery et al. 2023), revealing that sycophancy emerges from specific neural activity patterns that can be identified and modified.

While these interventions show sycophancy can be controlled through targeted modifications, fundamental questions remain about how models process conflicting information when user opinions contradict learned knowledge. Our work seeks to understand the information flow dynamics that cause sycophantic behavior in the first place.

Mechanistic Interpretability. Mechanistic interpretability (MI) aims to reverse-engineer neural networks into human-interpretable algorithms, moving beyond traditional explainable AI focused on input-output relationships (Zhang et al. 2025c; Hu et al. 2024; Bereska and Gavves 2024; Kästner and Crook 2024; Yao et al. 2025; Su et al. 2025; Zhang, Hu, and Wang 2025; Wang et al. 2025; Zhang et al. 2025b; Dong et al. 2025; Yang et al. 2025). Key techniques for transformers include logit-lens analysis (nostalgebraist 2020), which extracts meaningful token predictions from intermediate layers, revealing what the model "believes" after each step and how these distributions converge toward the final output (Stolfo, Belinkov, and Sachan 2023; Zhang et al. 2025d). Activation patching provides a causal perspective by substituting activations between inputs to identify which components are necessary and sufficient for specific behaviors (Wang et al. 2022; Zhang et al. 2024; Hong et al. 2024).

Recent applications of these MI techniques to sycophantic behavior have begun to illuminate the internal mechanisms involved, though with limitations. Yu, Merullo, and Pavlick (2023) used head attribution to identify attention heads that resolve conflicts between memorized facts and contradictory contextual information, but their findings on factual recall (e.g., world capitals) show limited generalizability across different knowledge domains. Similarly, steering vector approaches have identified specific activation space directions corresponding to sycophantic versus truthful responding and can successfully modify model behavior through targeted interventions (Panickssery et al. 2023), but these methods focus on controlling sycophancy rather than explaining why these particular directions emerge or what computational processes give rise to the observed activation patterns.

User Opinion Induces Sycophancy

Following previous studies, "sycophancy" in LLMs is defined as the model's tendency to conform to a user's explicitly stated opinion, even when that opinion is incorrect (Denison et al. 2024). To understand how models process conflicting information when user opinions contradict learned knowledge, we designed a simple experimental framework that avoids multi-stage benchmark complexity (Fanous et al. 2025). Prior work shows that incorrect user opinions reliably trigger sycophancy (Sharma et al. 2023), so we use this as our primary sycophancy trigger across seven model families.

Experimental Setup

Models. We evaluate seven LLMs from different families: *Llama3.1 8B-Instruct* (Dubey et al. 2024), *Qwen2.5 7B-Instruct* (Team 2024a,b), *OPT 6.7B* (Zhang et al. 2022), *Mistral 7B-Instruct v0.3* (Jiang et al. 2023), *Falcon 7B* (Technology Innovation Institute 2023), *OLMoE 1B-7B Instruct* (Muennighoff et al. 2024), and *Pythia 6.9B* (Biderman et al. 2023), all of comparable parameter size, ensuring observed differences are not due to scale effects.

Dataset. We use the Massive Multitask Language Understanding (MMLU) benchmark (Hendrycks et al. 2020). Its multiple-choice format provides clear ground-truth labels, while its 57 academic subjects ensure broad generalizability across knowledge domains. In our case, sycophancy occurs when the model selects the same answer choice that the user claims to believe, which we designed to always be incorrect.

Sycophantic Pressure Conditions. Prior work shows LLMs are sensitive to user opinions, but it remains unclear whether sycophancy is triggered simply by opinion presence or depends on perceived user expertise as well. To disentangle these factors, we vary how user opinions are presented. Brief examples are in Figure 1 with more in the Appendix.

- **Plain:** The model receives the original MMLU question with the instruction "Please only respond with a capital letter: A, B, C, or D." This format simplifies evaluation and avoids subjective LLM-as-a-judge assessments. This serves as a baseline to measure accuracy without sycophantic pressure.
- **Opinion-only:** To simulate sycophantic pressure, we prepend each question with a user opinion (e.g., "I believe the right answer is B") and the same response instruction. The user's opinion always corresponds to one of the three incorrect choices, selected uniformly at random. This tests whether simple opinion expression can induce models to abandon their learned knowledge.
- **Opinion with Expertise (First-person Credible Source):** To test if user authority affects sycophancy, we add expertise-level conditions that build on the `Opinion-only` setting. Users self-identify as *Beginner*, *Intermediate*, or *Advanced* (e.g., "I am a professor in computer science, and I believe..." for *Advanced*). Comparing sycophancy rates with `Opinion-only` setting allows us to measure two potential drivers: **opinion-driven** sycophancy (models conform simply because users express opinions) versus **authority-driven** sycophancy (models are additionally influenced by perceived user credibility). We refer to this as `First-pov`.

Evaluation Metric. For each sample, we log the model's selected answer, and then compute three metrics: (1) ***sycophancy rate*, or *agreement rate*** (Malmqvist 2024), the proportion of samples where the model agrees with the user's belief; (2) ***accuracy***, the proportion where it selects the correct answer; and (3) ***independent error rate***, the proportion of incorrect answers that disagree with both the user's belief and the ground truth, indicating autonomous errors.

Experimental Results

User Opinions Strongly Induce Sycophancy. Figure 2 demonstrates that when models are exposed to user opinions, their agreement rate with incorrect beliefs rises sharply, averaging 63.7% across all models, with a range from 46.6% to 95.1%. This highlights that even a simple, unsupported opinion is sufficient to substantially shift model predictions toward user agreement.

Expertise Framing Has Minimal Impact. From Figure 3, we can see sycophancy rates remain nearly unchanged